

Authentication Challenges in Customer Service Settings Experienced by Deaf and Hard of Hearing People

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ABSTRACT

Customer services are important for answering questions, providing information, and handling issues. Often people want to connect with a customer service representative, yet, this can be an accessibility barrier for Deaf and Hard of Hearing People (DHH) when voice calls are the only method. However, little is known about the challenges experienced by DHH people in solely voice-based customer service settings (e.g., authenticating themselves over the phone to their banks). To address this, we interviewed 18 DHH people to understand the challenges when remotely authenticating themselves with customer service. Though DHH people are not unique in their attitudes and behaviors toward mobile authentication, we found that voice-based authentication and services are challenging. Furthermore, DHH people can often be put in positions where they must trust third-party support (e.g., human interpreters) when discussing sensitive information to authenticate themselves through voice-based authentication and services. We present several avenues of future work needed to address these challenges.

CCS CONCEPTS

• **Human-centered computing** → **Accessibility**.

KEYWORDS

Customer Service, Deaf, Hard of Hearing, Video Relay Services

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1 INTRODUCTION

Customer service is defined by Salesforce as the “*delivery of proactive and immediate support to customers anytime*” [1]. Various industries provide customer services (e.g., hotels, internet retail, television, and media players) and must safeguard their user’s Personally Identifiable Information (PII). As a result, users are often required to authenticate themselves to proceed further. Currently, most companies support their customers through voice-based means, such as telephone calls, the fastest and sometimes the only means to resolve issues. However, phone calls can be inaccessible to Deaf and Hard of Hearing (DHH) people, thus phone call authentication poses a significant issue since there are about 466 million people (432 million adults and 34 million children) worldwide who have hearing loss ranging from mild to profound [29], and the percentage of people with hearing loss increases with age [27].

DHH people sometimes use interpreters and Video Relay Services (VRS) to authenticate themselves when facing voice-only services. VRS allows DHH people to connect to a sign language interpreter over video and continue conversing with the customer service agent. VRS makes the call a three-way conversation and can be available as standalone devices or mobile application services [28]. VRS has been used for many decades in healthcare, banking, and other domains as it is a federally approved service with strict confidentiality guidelines for interpreters [11].

However, third-party services are not always available. An auto-ethnographic report of the travel experience of a hard of hearing person illustrated the issue when InnoCaption was unavailable when traveling in another country and he needed a new bank card [16]. His bank did not provide an alternative to voice-based authentication, and he had to ask someone for help to authenticate himself on the phone to obtain a new card. This resulted in sharing personal information with a ‘middle man’ as that was the only option in the urgent situation [16]. Other situations, such as the COVID-19 pandemic [9], have increased the need for people to stay home, thus making it more likely that people are required to complete authentication and communication with customer service through telephones rather than in person. Unfortunately, the authentication practices of DHH people have not been adequately explored in prior research [4], especially when inaccessible systems like voice-based authentication exists [17].

Our work is timely and paramount to ensuring authentication in customer service settings is accessible. Therefore, we aimed to understand DHH people's customer service experiences during authentication. We interviewed 18 DHH individuals to understand their experience remotely authenticating themselves with customer service. However, we did not seek to compare the differences between DHH and hearing participants as this risks reinforcing normative, ableist beliefs [25].

We found that authentication in customer service settings can be accessible to DHH people so long as the authentication is textual or biometric and not voice-based. Still, authentication becomes more complex when additional mechanisms are required. First, DHH people may need to go outside of their comfort zone to ask and put their trust in others. Second, less accessible authentication is often long and tiresome, especially when customer service representatives are not accommodating of the other stakeholders helping (e.g., unfamiliarity with VRS setups involving interpreters). Therefore, it is essential that all people have sufficient support through the help of remote authentication in maintaining their right to privacy and security [7, 15]. Based on our findings, we discuss recommendations around addressing current remote authentication procedures with customer service (e.g., when DHH individuals have to use an interpreter while discussing sensitive information like healthcare) and opportunities to include and design new systems.

Paper Contributions: This paper makes two contributions. 1) a thematic analysis of 18 interviews with DHH participants on challenges with authentication in customer service settings. 2) a discussion on where future research efforts should focus on creating accessible mobile customer service authentication.

2 RELATED WORK

Little authentication research focuses on DHH people [4], especially in customer service settings. If customer service is limited to voice-based communication, like phone calls to receive a quick solution, it can create accessibility barriers for DHH users.

2.1 Obstacles During Authentication in Current Customer Service Settings for DHH Individuals

Our work focuses on DHH users of mobile devices. People who are deaf and Deaf (uppercase 'D' refers to their association with the Deaf community) have profound hearing losses, significantly reducing their hearing ability. People who are hard of hearing have mild, moderate, or severe hearing loss, which, in most cases, can be accommodated by hearing aids and cochlear implants [10]. Some DHH users prefer to use spoken language, while some prefer to use sign language to communicate [29]. Close to 466 million people around the world have a hearing impairment that ranges from mild to profound [29]. However, there is little to no work in authentication modes for DHH users [4]. Some work looked at extending the mobile authentication modes primarily designed to help vision impaired users to DHH users in the future [5, 12]. However, having other assistive devices for authentication, apart from the mainstream devices (e.g., iPhone), may make users self-conscious [37]. There needs to be a proper understanding of their challenges to design beneficial authentication modes.

Usually, privacy and security mechanisms are designed with the general population in mind and, at times, are poorly designed [2]. Such mechanisms are even less usable for people with disabilities if not made to be accessible [34]. As such, accessibility in privacy and security should be equally provided to everyone and must be addressed [7, 15]. We need more research exploring different types of disabilities and impairments for authentication to truly be accessible [38, 39].

In our modern world, we have to authenticate ourselves to access online banks [20], airlines, hotels, rental cars [14], and many other accounts. The differences mentioned above in hearing and communication have the potential to come between a person's privacy and security when authentication is only possible through voice [16]. DHH users may face authentication issues when it is required through voice, either to speak into the phone or to listen to a code for two-factor authentication without textual options. However, sometimes voicing is the only means of authentication with no other alternatives, such as for financial institutions, and that can hinder some DHH users [16]. While research has focused on DHH mobile authentication [4], it does not provide a complete account of other possible types of mobile authentication, which includes mobile/remote authentication in customer service settings through phone calls.

2.2 Video Relay Services Used During Authentication in Current Customer Service Settings for DHH Individuals

To communicate with customer service representatives, DHH users turn to assistive technologies like VRS for help. VRS would provide an interpreter allowing DHH people to sign with them and, in turn, communicate with hearing people over the phone on behalf of the DHH person [30], allowing them to communicate with banks [3], etc. Regarding banking, though customer service is claimed to be the essential duty [32], it is disappointing that only a few customer service representatives are aware of and accommodates using VRS [26] without them making the process even longer and tiresome. It is also worth noting the challenges that arose amidst the COVID-19 pandemic with inaccessible teleconferencing [21]. Preventive measures like lockdowns and social distancing made assistive technology like VRS much more of a relied-upon service. Increased reliance on VRS in the now remote world increases the privacy and security risks for DHH users as third parties are involved in the interaction. For example, most banking and its authentication were done via mobile devices, leading to the possibility of mobile banking becoming the future [36].

2.3 Summary and Research Question

Our literature review demonstrates that DHH users' challenges with authentication in customer service settings have not been investigated. Our qualitative research sought to understand DHH people's experiences with authentication in customer service settings. The findings of our work will lead to the creation of more accessible customer service authentication processes. Our study is centered around the research question: *What challenges do DHH users face during authentication in current customer service settings?*

3 METHODS

We distributed an online questionnaire to recruit people for interviews to understand the concerns DHH people had with authenticating themselves to mobile customer service representatives. Initially, our research was a more extensive project on DHH people's challenges and concerns with mobile authentication with customer service. However, the work we are reporting is focused on the customer service-specific part of our findings.

3.1 Online Interviews

Materials and Procedure: We ran semi-structured interviews remotely over Zoom with 18 participants covering several broad topics: demographic information, use of mobile devices for authentication, situational impairments related to authentication during the COVID-19 pandemic, and how current authentication processes can be improved. We advertised over four weeks among DHH online communities (e.g., Facebook and Reddit) and sought moderator approval when necessary. We determined that our interviews would take up to 45 minutes to discuss DHH people's concerns with the help of an ASL interpreter. All the questions were optional, and all the participants were reimbursed \$20.

Participants: We recruited 13 Deaf and five Hard of Hearing participants (Female = 8, Male = 9, Non-binary = 1) aged between 18–34 years. Eighteen DHH participants participated, with one previously completing the questionnaire (see Appendix A). Seven used hearing aids, eight used cochlear implants, and three used no assistive hearing devices. Fifteen of our participants used their phones over 3 hours a day, two indicating 1–2 hours per day and one indicating 0–1 hour per day.

Analysis: We followed the thematic analysis steps outlined by Braun and Clarke [6] to code the interviews. The lead author re-familiarized herself with the data after running the interviews and wrote accurate transcripts using the recordings. Initial codes were generated by reading through the data and were later discussed with our research team to generate final codes. Our team comprised four researchers; two have a background in human-computer interaction, and two in computer security. Our research team used Miro (<https://miro.com>) to support a remote, collaborative, and iterative discussion for organizing, collating, and collapsing codes (e.g., “positive attitudes to PIN/passwords” and “positive attitudes to multi-factor authentication” formed a higher level category “possible alternatives to customer service authentication”). This took several rounds, progressing from initial themes to final themes.

4 FINDINGS

We identified two general themes: 1) *bad experiences with remote authentication through customer service*, and 2) *good experiences with remote authentication through customer service*. We identified several subthemes, which we will discuss using supporting quotes.

4.1 Bad Experiences with Remote Authentication Through Customer Service

Our interviews uncovered situations with remote authentication through customer service that appear unique to DHH people. In some cases, DHH people are forced to call customer service with

no other option provided to authenticate themselves to their banks when they forget their passwords. For example:

P2: “I forgot my password for the account. So my account was just locked out. I considered that one of the worst experiences I’ve had because they didn’t have another way of communicating with the bank without calling them and then like the wait for calling the bank is 35 minutes [...] I don’t even want to like talk to them because I can’t really understand what they are saying most of the time [...] I like to do everything quickly and independently [...] if security is really a top priority, why do we have to call someone within the company, like customer experience, like when you can have an authenticator?”

Without other text-based options to recall forgotten passwords, DHH people are forced to seek help from interpreters. Interpreters are reached out to through VRS.

Video Relay Services: Ten participants mentioned using various VRS that help DHH people communicate with hearing people through the phone with the help of an ASL interpreter and video [11]. Participants used ZVRS, Z5, Z phone, Sorenson, Convo, and other videophones to authenticate themselves to their banks and other accounts when voice-over is the quickest and, in some cases, the only option available. Although online customer service support (e.g., chatbots) continues to grow, it can still be limited to interactions over the phone. If a third-party or interpreter has to mediate this, there are negative consequences as a result (e.g., the burden on DHH people finding another person, having to share details they may not wish to, thus compromising on their privacy and security, and experiencing issues where the service refuses to work with a third-party mediating). Calling the bank’s customer service over the phone removes DHH people’s independence (a critical issue) because they rely on interpreters through VRS. They shared their process of making a customer service call with the help of VRS. For example:

P2: “...my bank account locked, I will set up time on the Z phone and then I will set a time. Now I have to call them and inform them that [I] have an interpreter, and then the bank teller just goes through the same process as a hearing person would [...] I think [what] would be better for deaf and hard of hearing people is making like a much more independent software.”

When locked out of their bank accounts, DHH people are forced to rely on VRS and their interpreters to authenticate themselves in customer service settings. P2 also mentions the amount of time lost through the long three-way call.

Customer Service Unaware of VRS: Most customer service representatives are not used to a third-party being involved in conversations where sensitive information is discussed. Nevertheless, at most times, calling is the only option for a quick response. However, it is important to note that not all people possess video phones and cannot always authenticate themselves over the phone. For example:

P1: “A friend of mine, she’s profoundly deaf and her— she didn’t have a video phone at the time. And so she was trying to set up online banking, and her mom called for her. And

they like refused to talk to her and she was like explaining that you know, I'm deaf and then just finally then she went on to a video phone [...] and they still wouldn't talk to her because they were signing through an interpreter on the video phone and so the bank still wasn't like, allowing it, which is very common. Very annoying."

DHH people are sometimes deprived of the ability to authenticate themselves independently because their only option for communication with support is over the phone. However, P1's mother was not allowed to speak on her behalf. Customer service needs to accommodate DHH people.

Authentication is required when communicating with banks over the phone, and DHH people are forced to rely on assistive technology like VRS. Though having an interpreter through VRS helps, problems arise when the customer service representatives are unaware of VRS and the interpreter's gender differs from the DHH person trying to authenticate themselves. For example:

P18: *"If I call and there's a male interpreter, and the other person gets confused and says, Oh, your name says [NAME], you're female. But why am I hearing a male voice? So, that can sometimes get in the way [...] once someone was really resistant to it. But it was interesting because it's actually about my water bill from my house. So it wasn't even the bank [...] The person that I was talking to, the customer service person, I was trying to set up auto-pay, but the form wasn't available online, so they had to email it to me or something like that. So that I could set up auto-pay, but they, I guess, just didn't like that they were hearing a male voice."*

As P18 was someone in a region with a large DHH population, they expected the customer service representative to be more aware and accommodating of the use of VRS and interpreters. Calling customer service presents a noticeable inequitable imbalance to DHH people as it is unlikely that hearing individuals would find themselves in the same situation. For example:

P1: *"A lot of times, like with my boyfriend he doesn't, generally like talking on the phone [...] We have to go through a whole list of, you know, approved, like identifying him, making sure he is [name of boyfriend], and then, then they say like okay do you have permission to talk on the phone for you and then you know it's just like a longer process than it would be, if he just called himself."*

Too Much Trust Invested: DHH people tend to trust their family and friends, and, in most cases, third-party interpreters with their PII for assistance with customer service authentication, which is not something a hearing person likely has to do. Though DHH people consent and interpreters are required to keep the discussed PII confidential, it is risky. For example:

P12: *"Yeah, sometimes obviously, with the bank, of course, you have to authenticate yourself with your account number and your social security number, of course. But typically, the person in customer service will ask me like, they'll ask the interpreter if it's alright to share that information. Or sometimes, like obviously, for their job, it's confidential; they can't share any of that personal information. So like,*

sometimes, the customer service rep will just clear it with the interpreter and make sure that they're a trustworthy person before they get my personal information."

The DHH people depend on interpreters to make phone calls and do not question their PII's misuse. However, for hearing people, it is very natural to be extra careful with their PII and wonder if there are possibilities of misuse.

Extra Time Taken: Having to ask an interpreter to help with the calls makes the process longer (because of interpretation) and more frustrating (when misinterpreted). For example:

P8: *"You try to order something, so they will tell you press this number or press that number. Or, you know, but there, but I'm a deaf person. So I have to use an interpreter through that process so it adds the time that's needed to navigate those phone trees."*

P8 shared the challenges experienced as a DHH person with a voice-based phone call and the need for an interpreter to move further. As interpreters have to fingerspell each option, the whole process takes longer than it has to.

Help Needed from Strangers: DHH people are put in situations where they are sometimes forced to put their trust in strangers to receive help with their authentication. When the only option for authentication is via a telephone call, DHH people require help from bystanders to authenticate themselves. For example:

P18: *"I had to ask someone near me if they could listen to the call to get the code so that I could log in [...] But it definitely was uncomfortable to ask somebody. But at that time, I didn't really have a choice. You know, I had to use the number that they were calling me with."*

DHH people not only give up their independence of authenticating by themselves but are also pushed out of their comfort zones as they are required to place their trust in assistive technology like VRS and other people around them for the sake of authentication by audio means. Authentication should always have options for users without pushing people to rely solely on one mode. It is also crucial for customer service representatives to be aware of assistive technology like VRS to be patient and helpful.

4.2 Good Experiences with Remote Authentication Through Customer Service

Customer service experiences are sometimes good. Six participants mentioned having good experiences with customer service on call and two on the chat window. For example:

P13: *"I think occasionally I will, maybe, struggle with hearing the person like what they're asking, or sometimes I will ask them to repeat the question, and then they'll proceed pretty well."*

It is difficult for DHH people to communicate with hearing people over the phone, so it is important that hearing people are patient and accommodating. Which will improve the customer service experience.

Ideas to Improve Remote Authentication: When only phone calls are available to contact customer support, DHH people are forced to involve and trust a third-party interpreter (which is a long process in itself). DHH people discussed the importance of

alternative options to phone calls for increased independence. For example:

P15: *"I haven't run into the issue myself, I wouldn't doubt that there are some times when the only option is a phone call. And that's obviously not preferable for a variety of reasons [...] provide the ability for someone to call, someone to text, someone to use the computer, the chat box"*

To overcome their challenges with customer service voice authentication, we found participants requested textual communication and authentication with a customer service representative. At the moment, calling customer service is the most popular and quick method of resolving issues. However, being able to chat with customer service representatives (e.g., chatbox, email) will eliminate the middleman, giving DHH people complete control over their PII. Hence, it is crucial to include other authentication options without solely relying on one (currently, the phone call).

5 DISCUSSION

Our work contributes novel data outlining the challenges DHH people face during remote authentication through customer service. We conducted online interviews with 18 DHH people to discuss remote authentication. We uncovered situations where DHH people have to ask others for help when authenticating themselves over the phone. We found that DHH people can be forced to rely on assistive technology like VRS to communicate with someone who can assist them in the authentication process. Though DHH people trust the interpreters with their PII and maintain a professional relationship with them [13], having a third-party involved during the authentication process still poses privacy and security risks. Due to the lack of various customer-service authentication options, DHH people must trust others to aid the process, especially at times like this (COVID-19). Next, we reflect on our findings to make recommendations for future research directions.

5.1 Recommendations and Future Work

We highlight DHH people's issues with authenticating themselves over the phone. DHH people tend to ask interpreters from VRS to help them with voice-based authentication systems in the absence of friends and family. Furthermore, there are situations when customer service representatives are unaware of assistive technology like VRS and the challenges DHH people face in authenticating themselves over the phone in the absence of online chat and other alternatives.

5.1.1 The Need To Address Current Procedures. First, mobile customer services are not adequately considering the needs of their user base (in this case, DHH people). It is essential that companies include textual options [8] and not rely solely on voice-based communication. Companies should provide DHH people with accessible communication options (e.g., online chat). There is also an opportunity for the CHI community to explore novel communication systems specifically designed to support DHH people in customer service contexts. CHI has a history of researching customer service (e.g., [23, 24, 40]), and there is a need for the community to extend this focus on accessible customer services. It would also be beneficial to understand the needs of the companies that currently

do not offer alternative services and use co-design methods to create bespoke solutions that work for both the company and DHH customers.

Second, DHH people ultimately lose their independence when needing to rely on VRS technology or asking someone (e.g., family, friends, strangers) to help since an additional person enters the authentication process. DHH people are often compelled to trust people (including strangers) with their PII, thereby losing control over their data and compromising mobile authentication in customer service settings. Integrating alternative authentication methods for the PII part of the process could go some way toward minimizing risk. For example, when a DHH person uses VRS, they have access to an interpreter. Still, parts of the customer service procedure could instead be rerouted to an alternative device (e.g., sending a prompt to a phone to type in a Social Security Number rather than sharing through the interpreter).

Third, customer service representatives often do not accommodate interpreters from VRS, friends, or family, which makes the authentication process even more lengthy and cumbersome. The idea of increased time due to inaccessible services has been discussed in prior work and referred to as Crip Time [33], coined by Dr. Alston Kafer [18]. We envision new possibilities and alternatives to current authentication methods with a goal of reducing a time-related burden. Companies should train their customer representatives to be educated on VRS and its usage, which will go some way toward making more efficient interactions.

5.1.2 Opportunities To Design New Systems. Restructuring services could improve customer service authentication so that voice interaction is not the only means of resolving issues. For example, biometrics and offline One-Time-Passwords (OTP) are used to authenticate hotel booking and car rental systems [19, 22]. There is an opportunity for research in this area, and DHH people must be included in the design process. Incorporating text-based customer service solutions that include a secure PIN or password entry will allow DHH people to easily reach and authenticate themselves to customer service representatives. This will result in authentication methods that will benefit both DHH and hearing people. To move away from relying on third-party humans, researchers could also explore how to enable third-party AI-assistants on a DHH person's phone to communicate on their behalf with the customer service representative. For example, Google currently has automated call screening [31], where the AI-assistant answers the call and displays a transcript, so a person can see the conversation and decide if they want to pick up or not. A similar service could be explored, allowing DHH people to read the conversation and respond by typing a message, which the AI-assistant would then read aloud to the customer service representative. The benefit of such a solution would be that it provides a chat-based solution for DHH people when the company's customer service does not offer one.

5.2 Limitations

Although our sample is relatively large for accessibility research [35], the demographic details are skewed to a younger population and their experiences may not represent those of older DHH people. We also had to rely on different interpreters for each of the interviews. However, our interpreters are professional ASL signers who are

used to interpreting new situations. We are confident any difference were minimal due to participant responses being on topic and our similar findings between participants. Furthermore, we provided our interview guide to the ASL interpreters to ensure translation fidelity.

6 CONCLUSION

Many customer service protocols are limited to phone calls. This is problematic for DHH customers, who would benefit from alternative methods of communication. We interviewed 18 DHH people to understand their challenges with authentication in customer service settings. We found that authentication modes are accessible to DHH people so long as they are textual and not voice-based. We discuss recommendations around addressing current authentication procedures in customer service settings (e.g., DHH people use interpreters when discussing sensitive information, whereas hearing individuals do not need third-party human support). We make several recommendations and propose opportunities that include new authentication approaches for the CHI community to explore, which would allow DHH people to authenticate themselves with customer service more independently.

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A APPENDIX

Table 1: Interviewee demographics covering gender (M=Man; NB=Non-Binary; W=Woman), age range, hearing status (D=Deaf; HH=Hard of Hearing), use assistive hearing devices, and highest completed education

ID	Gender	Age	Hearing Status	Use of hearing devices	Highest completed education
P1	W	18-24	HH	Hearing aids	Some graduate school
P2	M	25-34	HH	Hearing aids	Associate degree
P3	NB	25-34	HH	Hearing aids	Undergraduate university
P4	W	25-34	D	Cochlear Implants	Undergraduate university
P5	W	25-34	D	Cochlear Implants	Graduate school
P6	W	18-24	D	None	Undergraduate university
P7	M	18-24	D	Hearing aids	Undergraduate university
P8	M	25-34	D	None	Undergraduate university
P9	M	18-24	D	Cochlear Implants	High school or equivalent
P10	W	18-24	D	Cochlear Implants	High school or equivalent
P11	M	25-34	D	Cochlear Implants	Undergraduate university
P12	M	18-24	D	Hearing aids	Associate degree
P13	W	18-24	D	Cochlear Implants	Associate degree
P14	M	18-24	HH	Cochlear Implants	High school or equivalent
P15	M	18-24	HH	Hearing aids	Some undergraduate university
P16	W	25-34	D	Cochlear Implants	Associate degree
P17	M	18-24	D	None	Undergraduate university
P18	W	25-34	D	Cochlear Implants	Graduate school